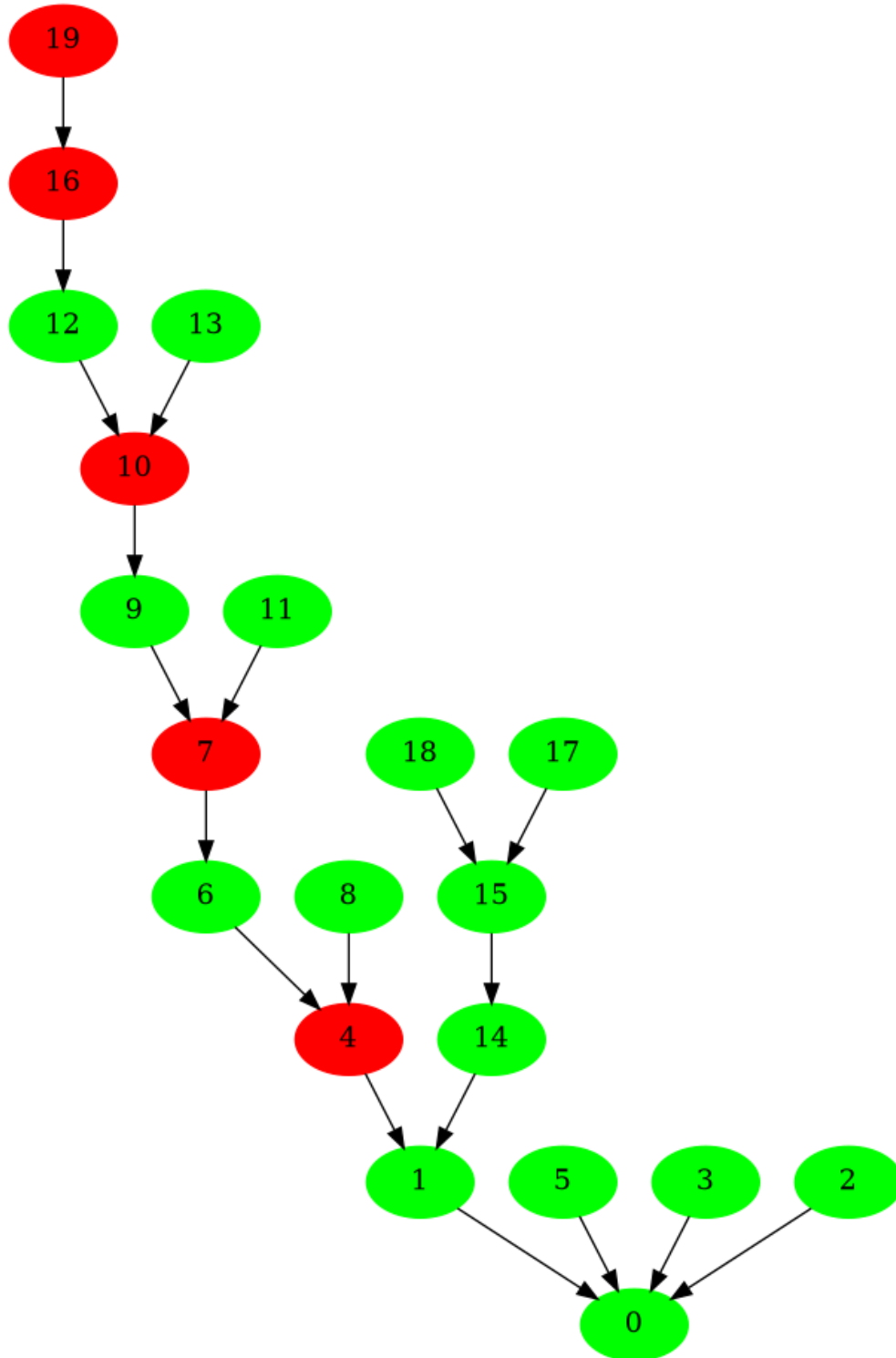


Blockchain: Assignment-2 Report

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Blockchain at Ringmaster at then end of simulation:



Analysis:

1) The ratio of the number of blocks generated by malicious nodes in the longest chain at the ringmaster to the total blocks in the longest chain at the ringmaster for various % malicious nodes in the network like 5,10,15, and so on with different timeout times(T_t).

Question-answers:

1. How does increasing the timeout time (T_t) affect block propagation in the presence of an eclipse attack?

→ In presence of an eclipse attack, increasing the timeout time will favour the honest chains. Due to eclipse attack, the honest blocks propagation is already slowed down, hence, increasing timeout time will increase the chances of receiving honest blocks of the honest nodes.

2. Suggest countermeasures to mitigate or weaken the above-described attack. [open-ended]

→ There are multiple counter-measures to this attack as following:

1) increase peers connected to one node. This will reduce the likelihood of eclipsing an honest node by all malicious nodes.

2) Instead of waiting for Timeout time for each block get request, honest nodes should send the block get requests to all nodes as soon as they get a hash. This will increase the network congestion, but will reduce the duration of hash and block receipt, hence increasing the speed of honest blocks propagation throughout the network.

3) Honest nodes could introduce some reputation system to reward honest blocks and punish malicious ones, and broadcast the reputation of each node time to time.